

LUCKBOUND

Master Game Design & Development Specification

Version 0.2 — Revised against a working prototype

Roll your fate. Enter the unknown. Become legendary.

What changed in v0.2

v0.1 described an intended game. v0.2 describes a game that runs. Phase 1 is complete and playtested in Roblox Studio, and this revision replaces speculation with measured values wherever the build has settled a question.

Three decisions materially changed the design: Fate became **true RNG** and no longer influences odds; onboarding became a **15-roll scripted arc** peaking on Epic; and the prototype gained a **fourth world**. Each is marked in place.

v0.1 remains authoritative on intent. Where the two disagree about what the game is *for*, v0.1 wins and this document is wrong. Where they disagree on implementation, v0.2 wins.

Source-of-truth scaffold for human collaborators and AI-assisted Roblox development.

1. Status

Phase 1 is complete and verified in Roblox Studio. A player spawns in the Crossroads, walks to the Fate Engine, presses ROLL, and receives a destination with its rarity revealed over a rarity-scaled animation.

Measure	Value
Server boot	9/9 steps, zero errors
Hub generation	5 zones, 436 instances, procedural at runtime
Automated tests	167 passing, headless, no Roblox required
Roll distribution	verified at 400,000 draws, worst drift 0.113pp
Worlds	5 defined (4 in the Phase 1 roll pool)
Source modules	32 Luau files

The North Star has been answered

§25 of v0.1 set the test: *after getting a rare item, does the player immediately want to roll again?*

Played with no combat, no loot, no audio and untextured blockout geometry, the owner's verdict was that **the rolls alone were fun**. Nothing was propping the loop up, which makes it the strongest available signal that the core premise holds.

2. Game Identity

Title: LUCKBOUND **Tagline:** Roll your fate. Enter the unknown. Become legendary.

Core fantasy: becoming increasingly capable of reaching rarer corners of reality — rarer destinations, stronger builds, stranger discoveries, and eventually global events.

Genre: RNG / adventure / light action RPG / collection / idle-retention.

Design rule: RNG determines possibility. Gameplay determines mastery.

3. Core Gameplay Loop

ROLL A WORLD → ENTER RANDOM DESTINATION → FIGHT & EXPLORE → DISCOVER RARE THINGS → ACQUIRE POWERS → DEFEAT BOSS → INCREASE FATE → RETURN TO HUB → ROLL AGAIN.

Three RNG layers: **World** — where am I? **Build** — what powers do I have? **Discovery** — what can I find?

Combat exists so the experience is not RNG → RNG → RNG. Idle progression is a retention layer, not the primary game.

Built so far: the roll, the reveal, the hub, and Fate progression. Everything from ENTER onward is Phase 2.

4. Fate & World RNG

REVISED IN v0.2 — Fate is true RNG

v0.1 §4 described Fate as granting agency over the roll: rerolls, destination choice, modifier control. **That is reversed.** No player state — Fate level, points, playtime or spend — changes the probability of any world. The draw is honest and auditable.

Two things this buys: the roll is **defensible** (every roll is logged with its seed, so “the game is rigged” has a verifiable answer), and a rare result **keeps its meaning** — a Mythic at high Fate is worth exactly what a Mythic at low Fate is worth.

The weighting code is retained and disabled behind one flag, so the decision is reversible.

What Fate is instead

Fate remains the primary meta-progression stat. It is a **score and unlock currency**: it gates features, titles, Hall of Fate standing and Fatebreak eligibility. What it does not do is sell better luck.

Open consequence, unresolved

With true RNG, a player at roll 200 faces **identical odds** to one at roll 16. “Stuck in Commons” is therefore a permanent condition rather than an early-game phase, and Fate cannot weight its way out of it.

The lever that stays consistent with true RNG: **Fate unlocks which pools you draw from, never how the draw resolves.** Higher Fate grants access to a pool with no Common in it. This preserves v0.1’s progression ladder as content access rather than odds tilting. Phase 3.

5. Onboarding — the first 15 rolls

New in v0.2. True RNG creates a problem: a new player may never see what the game is capable of. A scripted opening arc solves it without touching the odds.

Roll	World	Rarity	Roll	World	Rarity
1–2	Verdant Valley	Common	9–10	Verdant Valley	Common
3	Emberfall	Rare	11	Emberfall	Rare
4–5	Verdant Valley	Common	12–13	Verdant Valley	Common
6	Emberfall	Rare	14	Emberfall	Rare
7	Verdant Valley	Common	15	Verdant Valley	Common
8	Sky Citadel	EPIC	16+	true RNG	—

Two rules shape it

- **It never hands out a Mythic.** A guaranteed top tier devalues the top tier permanently, and the Fatebreak announcement depends on Mythic meaning something. The arc peaks on Epic; Astral Reach must be earned.
- **It approximates the real odds.** Counted out: 10 Common, 4 Rare, 1 Epic — roughly 67/27/7 against true RNG's 70/20/7/3 — and it ends on a Common. A front-loaded parade of rares followed by a wall of Commons would make roll 16 feel like a punishment.

The valleys are the point. Rares land at 3, 6, 11 and 14 with Commons between them, so each reads as a lift rather than an entitlement. Indexed by total rolls, so it survives a rejoin and cannot be farmed by disconnecting. Both rules are enforced by test.

6. Worlds

Revised in v0.2: the prototype carries four rollable worlds, not three. Sky Citadel (Epic) was added so onboarding could peak below Mythic.

World	Rarity	Weight	%	Boss	Biome
Verdant Valley	Common	7000	70%	The Rootbound Guardian	specified
Emberfall	Rare	2000	20%	Ignivar, the Molten King	specified
Sky Citadel	Epic	700	7%	—	none yet
Astral Reach	Mythic	300	3%	The Star Eater	specified
The Unknown	Unknown	5	Phase 3	—	none

Weights sum to 10000 so they read directly as percentages. Worlds carry an absolute weight and a phase gate; the pool renormalises over whatever is enabled, so Phase 3 can switch on the full eight-world table without retuning anything.

Known debt — Sky Citadel has no biome

Sky Citadel is live in the roll pool at 7% but is **data only**. The Biome Blueprint never drafted it, so its enemies, boss and loot tables are empty and its lighting values are invented rather than blueprint-sanctioned.

Phase 1 never enters a world, so this is sufficient today. **It must have a real biome before Phase 2 makes worlds enterable**, or rolling it sends a player somewhere that does not exist.

7. Rarity

Single source of truth for rarity across UI, portals, particles and loot glows. **Rarity colour is a UI and portal contract; biome palette is set dressing.** Emberfall's environment is fire-orange while its portal ring is Rare blue — theme colour must never bleed into rarity signalling. Enforced by test.

Rarity	Order	Hex	Reveal	Used by
Common	1	#9E9E9E	2.5 s	Verdant Valley

Rarity	Order	Hex	Reveal	Used by
Uncommon	2	#5FD97A	2.8 s	Ancient Ruins (Phase 3) — colour needs sign-off
Rare	3	#3E8EF7	3.2 s	Emberfall
Epic	4	#B24BF3	3.8 s	Sky Citadel
Legendary	5	#FFD447	4.5 s	Hall of Legends, Astral secondary
Mythic	6	#FF7A1A	5.5 s	Astral Reach
Unknown	7	#7CF5E0	7.0 s	Fatebreaks, The Unknown

Reveal duration scaling is the anticipation mechanic and the highest-leverage number in the game. Measured in play: 9.4 s between rolls for a Rare, 12.2 s for a Mythic. A rarer result demonstrably makes the player wait longer.

8. The Crossroads

Permanent hub, 120 studs across, generated procedurally at runtime from data. Roblox is Y-up, so the compass is the X/Z plane.

Zone	Anchor	Built
Fate Engine	(0, 0, 0)	PortalRig @1.5×, 24-stud marble platform, rune inlay, 6 orbiting shards, gold spotlight
Hall of Legends	north (0,0,−46)	3-tier amphitheatre, 5 World-First obelisks, cool museum spotlights
Discovery Archive	east (46,0,0)	Domed rotunda, 3 rings of orbiting shelves
Expedition Gate	south (0,0,46)	PortalRig @2×, 12-stud processional, torch pillars
Training Grounds	west (−46,0,0)	30×30 yard, 3 dummies + boss dummy (inert until Phase 2)
Global Observatory	up (0,40,0)	18-stud platform, glass orrery, spiral ramp

One shared **PortalRig** serves the Fate Engine (1.5×) and the Expedition Gate (2×), reskinned by rarity colour only and never duplicated per biome. It carries the **2.5-second spin-up** that must complete before the destination banner appears — enforced by test, because an instant reveal destroys the anticipation the whole loop rests on.

9. Architecture

Canonical principle: systems are reusable, content is data. A System contains behaviour and no content. Adding a world means editing one file under Content and changing no System. If implementing content requires changing a System, the schema is wrong.

Layer	Contents
Core	Types, Constants, GameConfig, Net, UITheme, Result
Pure cores	FateCore, ProgressionCore, ProfileSchema, EventCore — every decision, no Roblox globals
Util	WeightedRandom, Schema, PortalRig

Layer	Contents
Systems (server)	Save, Progression, Fate, Event, HubBuilder
Content	Worlds, Hub — plain tables
Client	StateController, ProximityController, HubEffects, FateRoll, GlobalAnnouncements

The pure-core split is why a 167-test suite exists. Decisions live in modules with no Roblox dependencies, so they run headlessly in CI; Systems hold only plumbing.

Boot order is fixed and content validation runs first — the server **refuses to start** on malformed content rather than serving it silently. Every boot step prints its own progress line, so a server that stalls names the step it stalled on.

Non-negotiable: the client sends no payload with a roll request; results come from a server-owned Random instance, never `math.random`; rolls are rate-limited and distance-checked server-side, silently; every roll is logged with its seed.

10. Late Joiners

New in v0.2. Anything globally visible must be visible to a player joining a server already in progress. A broadcast only reaches people already connected, so **server state is the source of truth and every broadcast is a side effect of changing it**. A player joining four minutes into a five-minute Fatebreak sees it with one minute left, not a fresh five.

Cross-server Fatebreaks preserve the original start time, so every server counts down to the same wall-clock end.

11. Roadmap

Phase	Scope	State
1 — Foundation	Hub, Fate Engine, roll, reveal, progression, save, client UI	COMPLETE , verified in Studio
2 — A game	Expedition entry, combat, enemies, boss, loot, inventory, Discovery Book, timer	next
3 — Compelling	Remaining worlds, modifiers, Fatebreaks, Fate pool unlocks, polish, audio	planned
Then	Stop. 10–50 testers. Watch whether they roll → smile → enter → discover → roll again.	—

Phase 1 deliberately excluded combat, loot, inventory, the Discovery Book, expedition entry, the timer, modifiers, AFK, trading and leaderboards. That exclusion list is what kept Phase 1 finishable, and the same discipline applies to Phase 2.

12. Open Decisions

#	Item	State
1	Blueprint §1.1's usage column contradicts §4.4/§5.4 on which world carries Mythic orange	Resolved toward §4.4/§5.4; needs confirmation
2	Rarity ordering — whether Legendary outranks Mythic	Kept Mythic above; needs confirmation
3	Uncommon has no blueprint-sanctioned colour	Prototype green retained
4	Environment colour vs rarity colour rule	Implemented; formal sign-off open
5	Expedition length — 720 s is likely too long for a tester	Revisit in Phase 2
6	Sky Citadel biome	Blocks Phase 2
7	Persistence unverified — DataStores need a published place	Blocks P1-8, P1-9

13. Canonical Terminology

Term	Meaning
The Crossroads	Permanent central hub
Fate Engine	Machine determining expedition destination
Fate	Meta-progression and unlock resource — does not influence odds
Expedition	Temporary access to a rolled world
Discovery	Collectible thing the player finds
Discovery Book	Collection and archive system
Fatebreak	Extraordinary event triggered by an exceptional discovery
Starfall	Example global event triggered by The Eternal Star
Ascension	Long-term prestige / endgame reset
World Rift	High-difficulty modifier-driven endgame challenge
PortalRig	Shared portal component, reskinned by rarity — new in v0.2
Onboarding arc	The scripted first 15 rolls — new in v0.2

Companion documents: *PROTOTYPE_BUILD_SPEC.md* (canonical architecture), *STATUS.md* (current state), *BLUEPRINT_RECONCILIATION.md* (open sign-offs), *TESTING.md*, *ART_DIRECTION.md*, *TOOLCHAIN_ACCESS.md*, *ADDENDUM_ASSET_PIPELINE.md*.